Graphic Era Hill University
DEHRADUN

IEEE
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Student Branch

SAARTHI'26 Submission

REWIND

SEMANTIC VIDEO SEARCH FOR CCTV

TEAM	TEAM LEADER	TEAM MEMBER	COLLEGE
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Problem Statement

CCTV cameras record every second around us. But when something goes wrong, like a theft, a hit-and-run or a missing person, **nobody can simply ask the footage what happened**. Someone has to sit and watch hours of video, hoping they don't blink at the wrong moment.

Ever wondered how police search CCTV when a crime happens? Even today, most of it is done by hand:

Manual playback
Someone watches hours of video on fast-forward

Timestamp guessing
Start at a witness's rough time

Motion search
Flags anything that moves, so busy roads flag everything

Split the hours
Footage is divided among several people to watch

NOT AN EXAGGERATION

200,000+

hours of CCTV footage
were reviewed by police after the 2011 London riots [1]

8–10 hours a day per officer

Does it get the job done? Mostly, but not completely. **Why?**

Tired eyes miss things
Hours of watching make key seconds easy to miss.

Too many cameras, too few people
Far more footage than people to review it.

To catch John, investigators don't need every hour of footage, only the right few seconds. What if they could just describe what they're looking for, and jump straight there?

Proposed Solution

Presenting to you all

REWIND

A semantic video search system for CCTV footage.

Describe what you saw. Get the clip.

YOU TYPE

“a white car stopping near the gate”

REWIND FINDS

CLIP (an AI model) matches your words to video frames

HNSW (a fast search index) finds the closest ones

YOU GET

#1 CAM-02 00:42:10–00:42:31

#2 CAM-01 01:17:48–01:18:02

Search by meaning
Describe it in plain words

Clips, not frames
5 s before and after each hit

Fast at scale
Own C++ index, built for speed

Evidence untouched
Original video is never changed

No face recognition
No tracking across cameras

Humans decide
It finds clips, people judge

Rewind doesn't decide who John is. It shows investigators exactly where to look first.

Technical Approach

PHASE 1 · INDEX THE FOOTAGE

done once, when videos are added

Import

Each video is locked read-only and fingerprinted (SHA-256), so the evidence never changes.

Sample

Take 1 frame every second. Skip near-identical frames, but never leave a gap over 10 s.

Embed

CLIP turns every kept frame into 512 numbers that describe what's in it.

Index

A fast C++ search graph (HNSW) stores them, each tagged with its camera and time.

PHASE 2 · ANSWER A SEARCH

every time someone types a query

Ask

Type a plain sentence like “white car near the gate”.

Understand

The same CLIP model turns your words into 512 numbers.

Match

HNSW finds the 200 closest frames without checking every one.

Group

Nearby hits (under 5 s apart) merge into one clip.

Review

You get ranked clips with 5 s of context, plus a timeline.

THE KEY IDEA

CLIP puts pictures and sentences in the same “meaning space”. A frame of a white car lands right next to the words “white car”, so searching video becomes **finding the nearest points.**

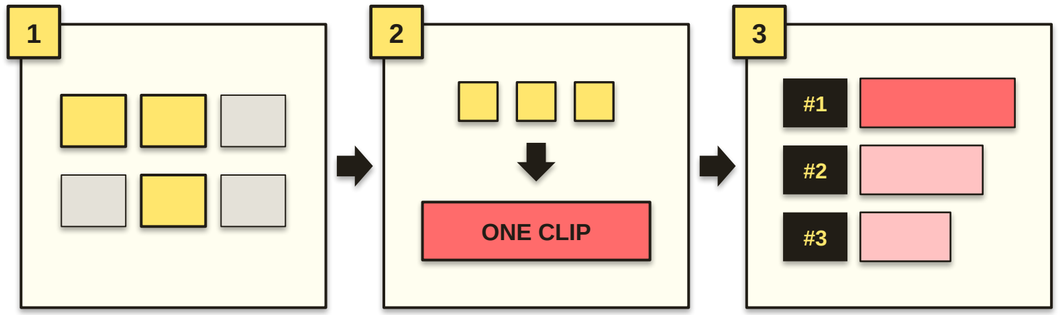
TECHNOLOGIES TO BE USED

PYTHON | Python | PyTorch | open_clip | OpenCV | NumPy

C++ | C++20 | CMake | Custom HNSW | cpp-httplib | GoogleTest

Detailed Description

HOW REWIND PICKS THE CLIPS



Spot frames that match your words — **Join nearby matches into one clip** — **Best clips first, with 5 s before and after**

Under the hood: top 200 frames → merge if ≤ 5 s apart → rank by best frame

HOW WE'LL PROVE IT WORKS

1. **Take 40+ real events**
   with known start and end times from the dataset's labels
2. **Describe each in plain English**
   e.g. “a white car stops near the gate”
3. **Lock half away as the final test**
   we never tune Rewind on it
4. **Compare with watching in order**
   also vs. random order, in daylight and at night

HEADLINE METRIC
How much video you must watch before the right clip shows up
(less is better)

DATA

Public CCTV dataset with labelled events
Candidate: VIRAT [5]
3–6 hours, day and night

ALSO MEASURED

Is the right clip in the top 1, 5 or 10?
Search speed, with and without HNSW

GUARDRAILS

Raw video read-only + SHA-256
No face recognition or tracking

Impact and Benefits

SOCIAL IMPACT

- Faster leads in thefts, hit-and-runs and missing-person cases
- Fewer key moments missed to tired eyes
- Privacy by design: no face recognition, no tracking

ECONOMIC IMPACT

- Works on existing cameras: no new hardware
- Runs on one machine, affordable for colleges and societies
- Frees staff hours for real follow-up work

ENVIRONMENTAL IMPACT

- **Modest, honestly: it is software, not green tech**
- Skips repeat frames, so less compute and storage
- Runs locally: no GPU cloud, no constant uploads

TARGET AUDIENCE

- Police and investigation teams
- Campus, hostel and housing-society security
- Traffic police, malls, parking lots and warehouses

Prototype

Already built:

Rewind reuses the engine from our image-search project

31,013
images indexed

0.946
top-10 recall vs exact

0.54ms
per search, 13.9× faster

11/11
test checkpoints passed

NEXT: THE 48-HOUR BUILD

- [ ] Import videos + sample frames
- [ ] Turn frames into CLIP numbers + index them
- [ ] Group matching frames into clips
- [ ] Test on held-out queries
- [ ] Web UI with clip playback

If time runs short, we cut UI polish, never testing.



WATCH OUR 2-MINUTE DEMO
youtu.be/caFGvhiDkv0

WATCH ON YOUTUBE

REWIND search your CCTV footage — *illustrative wireframe, not real results*

a white car stopping near the gate — SEARCH

Top 10 results · cameras: CAM-01, CAM-02

#1	CAM-02	00:42:10 – 00:42:31 match	Play
#2	CAM-01	01:17:48 – 01:18:02 match	Play
#3	CAM-02	03:05:12 – 03:05:20 match	Play

Timeline

CAM-01

CAM-02

Research & References

RESEARCH WORK

1. **City Security Magazine** – Met Police CCTV review after the 2011 London riots (Det. Supt. W. Lyle)
2. **Keval & Sasse** – “Not the Usual Suspects”: factors reducing CCTV effectiveness · Security Journal, 2010
3. **Radford et al.** – CLIP: Learning Transferable Visual Models From Natural Language Supervision · ICML 2021
4. **Malkov & Yashunin** – HNSW: approximate nearest-neighbour search with navigable small-world graphs · IEEE TPAMI 2020
5. **Oh et al.** – VIRAT: large-scale benchmark for event recognition in surveillance video · CVPR 2011

LINKS

Our engine github.com/ashutoshanthwall/semantic_image_retrieval_using_hnsw_indexing

[1] citysecuritymagazine.com/police-partnerships/cctv-advances-convictions-london-riots/

[2] discovery.ucl.ac.uk/id/eprint/19823/

[3] arxiv.org/abs/2103.00020

[4] arxiv.org/abs/1603.09320

[5] doi.org/10.1109/CVPR.2011.5995586